SDK - HTHardDll.dll Manual

VC++6.0 IDE

Note:

HTHardDll.dll was compiled under VC++6.0.
WORD: unsigned short
BOOL: bool
ULONG: unsigned long

The following ifdef block is the standard way of creating macros which make exporting from a DLL simpler. All files within this DLL are compiled with the DLL_API symbol defined on the command line. this symbol should not be defined on any project that uses this DLL. This way any other project whose source files include this file see DLL_API functions as being imported from a DLL, wheras this DLL sees symbols defined with this macro as being exported.

```
#ifndef    DLL_API
#define    DLL_API    extern "C" __declspec(dllimport)
#endif
```

Define __stdcall:

```
#define WIN_API __stdcall
```

Defines the struct for the application.

_HT_RELAY_CONTROL contains the relay control information.

```
typedef struct _HT_RELAY_CONTROL
{
    BOOL bCHEnable[4];
    WORD nCHVoltDIV[4];
    WORD nCHCoupling[4];
    BOOL bCHBWLimit[4];
    WORD nTrigSource;
    BOOL bTrigFilt;
    WORD nALT;
}RELAYCONTROL,*PRELAYCONTROL;
```

Parameters Remarks:

bCHEnable[4]: The channel is enable/disable.Value: 1 is enable; 0 is disable.
nCHVoltDIV[4]: The VOLT/DIV index of the channel.
nCHCoupling[4]: The coupling index of the channel .

Value：DC is 0；AC is 1；GND is 2；

bCHBWLimit[4]：The bandwidth of the channel.

Value: be enable is 1: be disable is 0.

nTrigSource：The index of the trigger source. If the channel has 4 channels, the value is : CH1 is 0, CH2 is 1, CH3 is 2 , ……

bTrigFilt：The High Frequency Rejection. Be enable is 1, disable is 0.

nALT：If the trigger is alternate trigger, the value is 1, otherwise is 0.

Example:

Declare a variable: RELAYCONTROL myRelayControl;

Declare a pointer: PRELAYCONTROL pRelayControl;

_HT_CONTROL_DATA contains some controls.

```
typedef struct _HT_CONTROL_DATA
{
    WORD nCHSet;//Channel Enable/Disable
                //The 0 bit: 0 is enable, 1 is disable.
                // The 1 bit: 0 is enable, 1 is disable.
                // The 2 bit: 0 is enable, 1 is disable.
                // The 3 bit: 0 is enable, 1 is disable.
    WORD nTimeDIV;          //The index of time Base
    WORD nTriggerSource;    // The index of the trigger source
    WORD nHTriggerPos;      //The Horizontal Trigger Pos (Value: 0 ~ 100)
    WORD nVTriggerPos;      //The Vertical Trigger Pos (Value: 0 ~ 255)
    WORD nTriggerSlope;     //The Edge trigger Slope (Rise slope is 0,Fall slope is 1)
    ULONG nBufferLen;       //The buffer Length
    ULONG nReadDataLen;     //The Data Length of have be read.
    WORD nALT;              // Is alternate trigger or not

}CONTROLDATA,*PCONTROLDATA;
```

Example:

Declare a variable: CONTROLDATA myControlData;

Declare a pointer: PCONTROLDATA pControlData;

Functions:

1. DLL_API BOOL WINAPI dsoSetUSBBus (

WORD nDeviceIndex

)

Return value:

If the function succeeds, the return value is TRUE.

If the function fails, the return value is FALSE.

Parameters:

nDeviceIndex

index of the device

Remarks:

If you change to USB communication, you must call this function firstly.

2. DLL_API WORD WINAPI dsoHTReadCalibrationData (

WORD nDeviceIndex,

WORD* pLevel,

WORD nLen

)

Return value:

If the function succeeds, the return value is1.

If the function fails, the return value is zero.

Parameters:

nDeviceIndex

index of the device

pLevel

pointer to buffer, save the device calibration data

nLen

the calibration buffer size

Remarks:

This function must be called when initializing the device.

3. DLL_API WORD WINAPI dsoHTSearchDevice(short* pDevInfo)

Return Value:

The number of have connected devices.

Parameters:

pDevInfo

pointer to devices information that have connected to PC.

Remarks:

You should call this function to know how many devices that have connected to PC. One PC supports 64 devices to be connected.

Example:

```
short DevInfo[64];
WORD nConnectedDevNum = 0;
//DevInfo Initial….
//…
//Call this function
nConnectedDevNum = dsoHTSearchDevice(DevInfo);
if (nConnectedDevNum > 0)
```

```
{
    For(int i=0 ;i<64 i++)
    {
        if (DevInfo[i] = = 0)
        {
            // has device. The device is not used.
        }
        else if (DevInfo[i] = = 1)
        {
            // has device. The device is used.
        }
        else
        {
            //No device was found.
        }
    }
}
```

4. DLL_API WORD WINAPI dsoHTSetCHPos(
WORD nDeviceIndex,
WORD* pLevel,
WORD nVoltDIV,
WORD nPos,
WORD nCH
)

Return Value:
If the function succeeds, the return value is zero.
If the function fails, the return value is zero.

Parameters:
nDeviceIndex
index of the device
pLevel
pointer to calibration level.
nVoltDIV
index of channel
nPos
the position of channel, range: 0 ~ 255 (8bit)
nCH
the target channel (CH1 is 0, CH2 is 1, CH3 is 2. …….)
Remarks:
Set channel level (Zero Level)
Example:

```
WORD nDeviceIndex = 0;
WORD CHLevel[256]; // the calibration level, see dsoHTReadCalibrationData function
WORD nVoltDIV = 6; //
WORD nPos = 128; //
WORD nCH = 0; //
//
if ( 0 = = dsoHTSetCHPos(nDeviceIndex,CHLevel,nVoltDIV,nPos,nCH) )
    ; //Fail
else
    ; //Succeed
```

5. DLL_API WORD WINAPI dsoHTSetVTriggerLevel (
WORD nDeviceIndex,
WORD* pLevel,
WORD nPos)

Return value:

If the function succeeds, the return value is nonzero.

If the function fails, the return value is zero.

Parameters:

nDeviceIndex

index of current device

pLevel

pointer to calibration level

nPos

the position of the trigger level, range : 0 ~ 255 (8bit)

Remarks:

Set the position of the trigger level.

Example:

```
WORD nDeviceIndex = 0;
WORD CHLevel[256]; //the calibration level, see dsoHTReadCalibrationData function
WORD nPos = 128;
//call function
if ( 0 = = dsoHTSetVTriggerLevel( nDeviceIndex, CHLevel, nPos) )
        ; //Fail
else
        ; //Succeed
```

6. DLL_API WORD WINAPI dsoHTSetHTriggerLength(
WORD nDeviceIndex,
ULONG nBufferLen,
WORD HTriggerPos,
WORD nTimeDIV，
,WORD nYTFormat

)

Return value:

If the function succeeds, the return value is nonzero.

If the function fails, the return value is zero.

Parameters:

nDeviceIndex

index of current device.

nBufferLen

the buffer length

HTriggerPos

the horizontal trigger position(percent), range : 0 ~ 100

nTimeDIV

index of the TIME/IDV

nYTFormat

the mode of the horizontal format. Value: 0: Normal, 1: Scan, 2: Roll

Remarks:

Set the trigger length

Example:

```
WORD nDeviceIndex = 0;
ULONG nBufferLen = 10240;//10K
WORD HTriggerPos = 50; //50%
WORD nTimeDIV = 12; //40us
//Call function
if ( 0 = = dsoHTSetHTriggerLength(nDeviceIndex,nBufferLen,HTriggerPos,
                                                    nTimeDIV) )
    ; //Fail
else
    ; //Succeed
```

7. DLL_API WORD WINAPI dsoHTSetCHAndTrigger(

WORD nDeviceIndex,

RELAYCONTROL RelayControl)

Return value:

If the function succeeds, the return value is nonzero.

If the function fails, the return value is zero.

Parameters:

nDeviceIndex

index of current device

RelayControl

See “_HT_RELAY_CONTROL”

Remarks

Set channel and trigger

Example:

```
WORD nDeviceIndex = 0;
RELAYCONTROL RelayControl;
//RelayControl Initial….
//Call function
if ( 0 = = dsoHTSetCHAndTrigger(nDeviceIndex,RelayControl) )
    ; //Fail
else
    ; //Succeed
```

8. DLL_API WORD WINAPI dsoHTSetCHAndTriggerVB(
WORD nDeviceIndex,
WORD* pCHEnable,
WORD* pCHVoltDIV,
WORD* pCHCoupling,
WORD* pCHBWLimit,
WORD nTriggerSource,
WORD nTriggerFilt,
WORD nALT
)

Return value:

If the function succeeds, the return value is nonzero.

If the function fails, the return value is zero.

Parameters:

nDeviceIndex

index of current device

pCHEnable

point to Channel Onoff Item of "_HT_RELAY_CONTROL"

pCHVoltDIV

point to Channel Voltate Item of "_HT_RELAY_CONTROL"

pCHCoupling

point to Channel Coupling Item of "_HT_RELAY_CONTROL"

pCHBWLimit

point to Channel BWLimt Item of "_HT_RELAY_CONTROL"

nTriggerSource

Trigger Source Item of "_HT_RELAY_CONTROL"

nTriggerFilt

tTrigger Filt Item of "_HT_RELAY_CONTROL"

nALT

Alt Item of "_HT_RELAY_CONTROL"

Remarks

Set channel and trigger，use for VB program environment

Example:

```
WORD nDeviceIndex = 0;
```

```
WORD pCHEnable[4] = {1,1,1,1};
WORD pCHVoltDIV[4] = {6,6,6,6};
WORD pCHCoupling[4] = {1,1,1,1};
WORD pCHBWLimit[4] = {0,0,0,0};
WORD nTriggerSource = 0; //CH1
WORD nTriggerFilt = 0;
WORD nALT = 0;
// Call function
if(0==dsoHTSetCHAndTriggerVB(nDeviceIndex,pCHEnable,pCHVoltDIV, pCHCoupling,
        pCHBWLimit, nTriggerSource, nTriggerFilt, nALT) )
        ; //Fail
    else
        ; //Succeed
```

9. DLL_API WORD WINAPI dsoHTSetTriggerAndSyncOutput(
WORD nDeviceIndex,
WORD nSynMode,
WORD nlaTriggerMode,
WORD nTriggerMode,
WORD nTriggerSlope,
WORD nPWCondition,
ULONG nPW,
USHORT nVideoStandard,
USHORT nVedioSyncSelect,
USHORT nVideoHsyncNumOption,
WORD nSync
)

Return value:

If the function succeeds, the value is nonzero.

If the function fails, the value is zero.

Parameters:

nDeviceIndex

index of current device

nSynMode,

type of Trigger Source

value:

0: when select CH /CH2 /EXT as Trigger Source.

5: when select D0~D15 as Trigger Source

nlaTriggerMode,

he mode of the trigger D0~D15: Edge trigger is 0, Pulse Trigger is 1.

nTriggerMode

the mode of the trigger. Edge trigger is 0, Pulse Trigger is 1.

nTriggerSlope

the slope of the edge trigger, Rise slope is 0, Fall slope is 1.

nPWCondition

the condition of the pulse trigger

value:

+ Less is 0

+ Equal is 1

+ More is 2

- Less is 3

- Equal is 4

- More is 5

nPW

the pulse width value. The time range of the pulse width is 10ns ~ 10s, the pulse width value range is 2 ~ 2000000000（10ns/5 ~ 10000000000ns/5）

nVideoStandard

Set video standard, value : 0 is PAL/SECAM ,1 is NTSC

nVedioSyncSelect

Set video sync, value:

All Lines：0

All Field：1

Odd Field：2

Even Field；3

Line Num：4

nVideoHsyncNumOption

Set video line number, the min value is 1.

nSync

The synchro output is enable or disable. 1 is enable, 0 is disable.

Remarks:

set trigger

Exmaple:

```
WORD nDeviceIndex = 0;
WORD nSynMode = 0;//Trigger CH1/CH2/EXT
WORD nlaTriggerMode = 0; / Edge Trigger
WORD nTriggerMode = 0; //Edge Trigger
WORD nTriggerSlope = 0; //Rise Slope
WORD nPWCondition = 0; //
ULONG nPW = 2; //10ns, nPW = 10ns/5。
WORD nSync = 0; //Closed
//Call function
if (0 = = dsoHTSetTriggerAndSyncOutput(nDeviceIndex, nSynMode,
                                       nlaTriggerMode ,nTriggerMode, nTriggerSlope,
                                       nTriggerCondition, nPW, nSync) )
    ; //Fail
else
    ; //succeed
```

10. DLL_API WORD WINAPI dsoHTSetSampleRate(
WORD nDeviceIndex,
WORD nTimeDIV,
WORD nYTFormat
)

Return value:

If the function succeeds, the return value is nonzero.

If the function fails, the return value is zero.

Parameters:

nDeviceIndex

index of the current device

nTimeDIV

index of the TIME/DIV

nYTFormat

the mode of horizontal format. 0: Normal, 1: Scan, 2: Roll

Remarks;

Set sampling rate

Exmaple:

```
WORD nDeviceIndex = 0;
WORD nTimeDIV = 12; //index of the TIME/IDV.
//Call function
if (0 = = dsoHTSetSampleRate(nDeviceIndex, nTimeDIV) )
    ; //Fail
else
    ; //Succeed
```

11. DLL_API WORD WINAPI dsoHTStartCollectData(WORD nDeviceIndex)

Return value:

If the function succeeds, the return value is nonzero.

If the function fails, the return value is zero.

Parameters:

nDeviceIndex

index of current device

Remarks:

Notify the device to prepare collecting data

Example:

```
WORD nDeviceIndex = 0;
if (0 = = dsoHTStartCollectData(nDeviceIndex) )
    ; //Fail
else
```

; //Succeed

12. DLL_API WORD WINAPI dsoHTStartTrigger(WORD nDeviceIndex)

Return value:
If the function succeeds, the return value is nonzero.
If the function fails, the return value is zero.
Parameters:
nDeviceIndex
index of the current device
Remarks:
Notify the device to detect trigger signal.
Example:

```
WORD nDeviceIndex = 0;
if (0 = = dsoHTStartTrigger(nDeviceIndex) )
    ; //Fail
else
    ; //Succeed
```

13. DLL_API WORD WINAPI dsoHTForceTrigger(WORD nDeviceIndex)

Return value:
If the function succeeds, the return value is nonzero.
If the function fails, the return value is zero.
Parameters:
nDeviceIndex
index of the device
Remarks:
Force to trigger the signal
Example:

```
WORD nDeviceIndex = 0;
if (0 = = dsoHTForceTrigger(nDeviceIndex) )
    ; //Fail
else
    ; //Succeed
```

14. DLL_API WORD WINAPI dsoHTGetState(WORD nDeviceIndex)

Return value:
If the device prepares to collect data, the return value is 1.
If the device is detecting trigger signal, the return value is 3.

If the device has collected data, the return value is 7.
You can read the data only the return value is 7.

Parameters:

nDeviceIndex

Index of the device

Remarks:

Retrieve the collect state of the device.

Example:

```
WORD nDeviceIndex = 0;
while ( dsoHTGeState(nDeviceIndex) != 0x07)
{
    continue;
}
//Read data from the device
```

15. DLL_API WORD WINAPI dsoHTGetData (

```
WORD nDeviceIndex,
WORD* pCH1Data,
WORD* pCH2Data,
WORD* pCH3Data,
WORD* pCH4Data,
PCONTROLDATA pControl
);
```

Return value:

If the function succeeds, the return value is nonzero.
If the function fails, the return value is zero.

Parameters:

nDeviceIndex

idex of the device

pCH1Data

pointer to CH1 Data buffer

pCH2Data

pointer to CH2 Data buffer

pCH3Data

pointer to CH3 Data buffer

pCH4Data

pointer to CH4 Data buffer

pControl

Set _HT_CONTROL_DATA

Remarks:

Read the data to PC buffer from the device.

Example:

```
WORD nDeviceIndex = 0;
WORD nInsertMode = 2; //Sinx /x
if ( 0 = = dsoSDGetData ( nDeviceIndex, pCH1Data, pCH2Data, pCH3Data, pCH4Data
                          , pControl) )
    ; //Fail
else
    ; //Succeed
```

16. DLL_API BOOL WINAPI laSetTrigSource(
WORD DeviceIndex,
unsigned char nSour
)

Return value:

If the function succeeds, the return value is true.

If the function fails, the return value is false.

Parameters:

nDeviceIndex

idex of the device

nSour

trigger logic channel

value:

0~15: D0~D15。

Remarks:

set logic D0~D15 as trigger source

Example:

```
WORD nDeviceIndex = 0;
unsigned char  nSour  =  0; // D0
if ( false = = laSetTrigSource(DeviceIndex, nSour))
    ; //Fail
else
    ; //Succeed
```

17. DLL_API BOOL WINAPI laSetVoltDiv(
WORD DeviceIndex,
WORD *nlaVoltDiv
)

Return value:

If the function succeeds, the return value is true.

If the function fails, the return value is false.

Parameters:

nDeviceIndex

idex of the device

nlaVoltDiv

point to Voltage of 16 logic channel，

value

0~119: -6V~+6V

Remarks:

set gate level of logic channel

Example:

```
WORD nDeviceIndex = 0;
WORD nlaVoltDiv[16];
WORD i;
for(i=0;i<16;i++)
{
    nlaVoltDiv[i] = 58;//-0.2V
}
if ( false = = aSetVoltDiv(WORD DeviceIndex, nlaVoltDiv))
    ; //Fail
else
    ; //Succeed
```

17. DLL_API BOOL WINAPI dsoGetDisplayDataFromSourceDataSingle(

short *pDisplayData,
WORD *pSourceData,
WORD *LeverPos,
ULONG nDataLen,
WORD nCH
)

Return value:

If the function succeeds, the return value is true.

If the function fails, the return value is false.

Parameters:

pDisplayData

point to data.

pSourceData,

point to data.

LeverPos,

point to display data.。

nDataLen

date length ,default:10240

nCH

one of 18 channel

value:

0~17

Remarks:

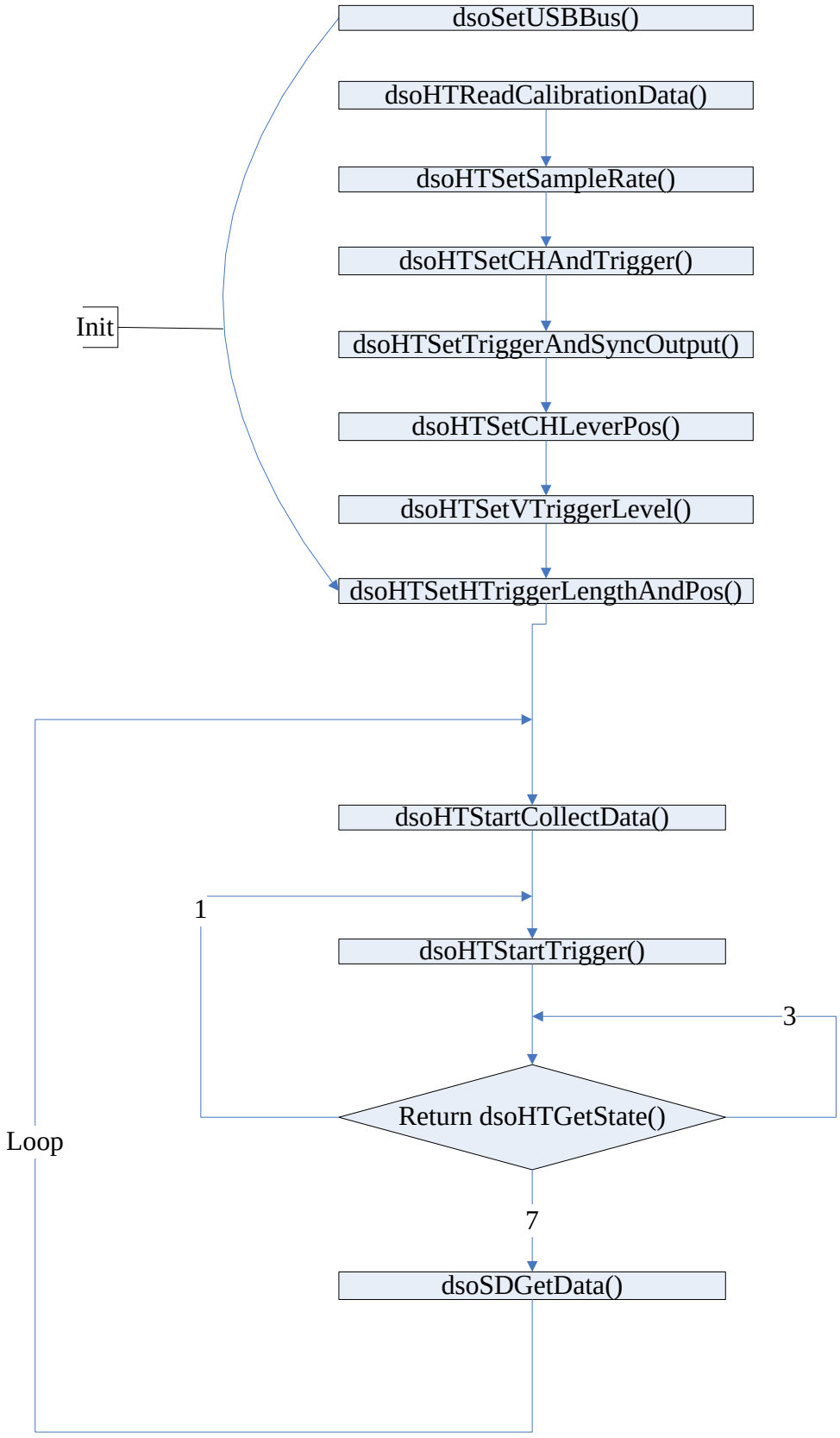
find out value of 18 channels from 4 channels collected.

CH1/CH2 unchanged,CH3 is D0~D7,CH4 is D8~D15

Example:

```
short pDisplayData[18][10240];    //value of 18 channels
WORD pSourceData[4][10240];   //4 channels collected
WORD LeverPos[18];                // display data of 18 channels
ULONG nDataLen = 10240;       //default
WORD nCH;                         //channel
    for (i=0;i<18;i++)
    {
        if(i<2)
        {
            if(!(dsoGetDisplayDataFromSourceDataSingle(pDisplayData[i],
                pSourceData[i] , LeverPos, nDataLen,i)))
            {
                //Error
                break;
            }
        }
        else if(i<10)
        {
            if(!(dsoGetDisplayDataFromSourceDataSingle(pDisplayData[i],
                pSourceData[2] , LeverPos, nDataLen,i)))
            {
                // Error
                break;
            }
        }
        else if(i<18)
        {
            if(!(dsoGetDisplayDataFromSourceDataSingle(pDisplayData[i],
                pSourceData[3] , LeverPos, nDataLen,i)))
            {
                // Error
                break;
            }
        }
    }
    if (i == 18)
    {
        //Succeed
    }
```

Flow-Chart:


dsoSetUSBBus()
dsoHTReadCalibrationData()
dsoHTSetSampleRate()
dsoHTSetCHAndTrigger()
Init
dsoHTSetTriggerAndSyncOutput()
dsoHTSetCHLeverPos()
dsoHTSetVTriggerLevel()
dsoHTSetHTriggerLengthAndPos()
dsoHTStartCollectData()
1
dsoHTStartTrigger()
3
Return dsoHTGetState()
Loop
7
dsoSDGetData()